

FINITENESS OF PLANAR FIVE-BODY CENTRAL CONFIGURATIONS FOR GENERIC EQUAL-PAIR MASSES

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ABSTRACT. Albouy and Kaloshin proved that the planar five-body problem has finitely many central configurations for all masses outside a codimension-two exceptional set. One component of that set is the equal-pair family $m_1 = m_2, m_3 = m_4$. We prove that for generic masses (a, a, b, b, c) in this family, that is, for all positive triples (a, b, c) outside a proper algebraic hypersurface, there are again only finitely many central configurations up to similarity. The proof is tropical. A polyhedral fan of 16,438 cones contains every possible asymptotic direction of an unbounded family of solutions. We rule out all but 98 cones with exact certificates: Laurent unit identities, explicit eliminations, and a modular criterion that checks a whole star of neighboring cones at once. The 98 remaining cones lie in one open half-space, which forces finiteness. Adding the fixed-potential equations and using the fact that only finitely many potential values occur turns this into a statement about physical configurations. Every certificate can be replayed with a standalone exact-arithmetic verifier.

1. INTRODUCTION

A configuration of n point masses $m_i > 0$ at positions $x_i \in \mathbb{R}^2$ is a *central configuration* (CC) if

$$\sum_{j \neq i} \frac{m_j (x_j - x_i)}{|x_j - x_i|^3} = -\lambda (x_i - x_{\text{cm}}) \quad \text{for all } i,$$

where x_{cm} is the center of mass and $\lambda > 0$. Smale's sixth problem asks whether, for every choice of positive masses, there are only finitely many CCs up to similarity. Hampton and Moeckel settled the case $n = 4$ [2]. Albouy and Kaloshin [1] settled $n = 5$ for all masses outside an explicit codimension-two subvariety $\mathcal{S}$ of the mass space. In Section 8.6, equation (32) of [1] is precisely $m_1 = m_2$ and $m_3 = m_4$. Their Theorem 6, proof part (ii), adds this locus and its relabelings to the exceptional set. Parts (i), (iii), and (iv) add other codimension-two mass loci; the equal-pair locus is not the entire positive-mass exceptional set. Our theorem concerns a relative-generic subset of this one family.

Jensen and Leykin [3] gave a tropical proof for generic masses in the ambient mass space. Their equal-pair valuation experiments in Section 4.2 do not yield pointed recession cones. Equal valuations allow unequal masses with the same leading order; here we impose the exact identities $m_1 = m_2$ and $m_3 = m_4$ in the coefficient field. The additional algebraic exclusions, rather than the support fan alone, remove the obstruction to pointedness.

Moczurad and Zgliczyński [5] certify counts at selected exceptional mass choices, including choices in the same equal-pair family, and discuss persistence in small neighborhoods. Their investigation of this family concerns specific mass choices rather than a relative Zariski-open subset of the family.

The all-equal planar case was already classified by Moczurad and Zgliczyński [7]; Lee and Santoprete [8] had earlier classified isolated equal-mass solutions. These results concern a special point of the mass family. Results on kite configurations [10], symmetric stacked configurations [11], and trapezoidal configurations with two equal mass pairs [12] impose geometric restrictions. The mass equalities in the present theorem do not impose reflection symmetry or a prescribed shape on the central configurations.

Hampton and Jensen [9] previously used polyhedral initial-ideal computations and elimination to prove spatial five-body finiteness with explicit mass exceptions. Chang and Chen [13, 14] develop symbolic diagram and mass-relation methods toward planar six-body finiteness; their combined preprint [15], Section 10.6, retains Diagram 19 for the mass relation $m_1 = m_2$, $m_4 = m_5$, equivalent to our family after relabeling. Those equalities already hold here, so they cannot exclude that diagram. Our method instead proves emptiness of distance-initial systems within the equal-pair family, using fixed-potential identities as additional equations; it does not identify individual support cones with their diagrams. Recent exact replication and tropical certificate screening by Santibáñez-Leal [16] concern ambient-generic results and valuation experiments. These works are precedents for the algebraic and verification techniques used here. Wang and Zhao [6, Theorem 8.1 and Corollary 8.2] obtain perturbative finiteness for non-scalar balanced configurations under a central-configuration finiteness hypothesis; their five-body corollary uses masses outside the Albouy–Kaloshin exceptional set.

Our contribution is a finiteness theorem for generic masses inside the equal-pair exceptional family. It covers all real planar configurations, with no symmetry or shape assumption. It does not settle every mass in the family, and it does not count the configurations.

Outline of the proof.

Suppose some mass choice had infinitely many central configurations. Then some distances would have to grow or shrink along a curve of solutions, and the leading exponents of that growth form a weight vector $w \in \mathbb{R}^{10}$. Tropical geometry says that such a w must lie in the tropical variety of the equations. That variety is contained in a computable fan Σ with 16,438 cones (Section 2). The proof has three steps.

- (1) *Exclude cones.* For each cone we show that the leading-order equations have no solution with all distances nonzero. This is done with explicit polynomial identities, eliminations, or a modular criterion (Sections 3–5). We exclude 16,340 cones.
- (2) *Separate the rest.* The remaining 98 cones all lie strictly on one side of a single hyperplane. A positive-dimensional tropical variety cannot lie in such an open half-space, so the solution variety is finite (Lemma 8).
- (3) *Return to physics.* The argument is applied to the equations with the potential fixed at a value u . Only finitely many values of u occur for real configurations (Lemma 9), so the finitely many solutions for each u give finitely many configurations overall.

Each certificate holds as an identity over the field $\mathbb{Q}(a, b)$. It therefore remains valid away from finitely many mass polynomials, and that is where the word “generic” enters.

Theorem 1. *There is a nonempty Zariski-open subset $\mathcal{O} \subset \{(a, b, c) \in \mathbb{R}_{>0}^3\}$ such that for every $(a, b, c) \in \mathcal{O}$, the planar five-body problem with masses (a, a, b, b, c) has finitely many central configurations up to similarity. Here similarity identifies translations, rotations, reflections, and positive changes of scale; body labels are retained.*

2. EQUATIONS AND TROPICAL SET-UP

The next section fixes the algebraic model used throughout. We encode a configuration by its mutual distances, retain the equations that enforce planarity, and then add the potential relation needed later to control scale.

We use the mutual-distance coordinates r_{ij} , $1 \leq i < j \leq 5$, in lexicographic pair order, with $r_{ji} = r_{ij}$ and $r_{ii} = 0$. Normalize the multiplier to $\lambda = M = \sum_i m_i$ and set $S_{ik} = r_{ik}^{-3} - 1$ for $i \neq k$. The ordered distance equations are

$$(1) \quad A_{ij} = \sum_{k \neq i} m_k S_{ik} (r_{jk}^2 - r_{ik}^2 - r_{ij}^2) = 0, \quad i \neq j.$$

There are $5 \cdot 4 = 20$ equations: the first index specifies the force vector being tested. We also use the ten redundant symmetric equations $A_{ij} + A_{ji} = 0$, $i < j$, together with the five determinants

$$(2) \quad \det \begin{pmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & r_{ij}^2 & r_{ik}^2 & r_{i\ell}^2 \\ 1 & r_{ij}^2 & 0 & r_{jk}^2 & r_{j\ell}^2 \\ 1 & r_{ik}^2 & r_{jk}^2 & 0 & r_{k\ell}^2 \\ 1 & r_{i\ell}^2 & r_{j\ell}^2 & r_{k\ell}^2 & 0 \end{pmatrix} = 0, \quad i < j < k < \ell.$$

We clear the denominators by multiplying A_{ij} by $\prod_{k \neq i} r_{ik}^3$ and each symmetric equation by the product of its two denominators. After removing common distance monomials and nonzero rational scalars, we obtain the stored system $F = (f_1, \dots, f_{35})$: first the twenty ordered equations, then the ten symmetric equations, and finally the determinants. The indices are lexicographic within each block. These removals are units on the distance torus, and no mass factor is removed. The formula comparison in `companion/checkers/verify_equations.py` reconstructs all thirty-five polynomials from (1)–(2) and matches `data/sys.json` exactly up to those units. The formulas agree with [3, Section 2.2]. Although the symmetric sums are algebraically redundant, their support hypersurfaces can strengthen the tropical prevariety.

Before turning to tropical geometry, we record why these scalar equations capture the real problem. This distinction matters because the algebraic variety considered later also contains complex torus points that need not represent physical configurations.

Lemma 2 (Physical meaning of the distance equations). *For positive masses and a collision-free real planar configuration, its positive mutual distances satisfy the twenty equations (1) if and only if the configuration is central with multiplier M . This includes collinear configurations. Every such distance tuple satisfies (2) and hence F .*

Proof. Put $q_i = \sum_{k \neq i} m_k S_{ik}(x_k - x_i)$. The cosine identity gives $A_{ij} = -2q_i \cdot (x_j - x_i)$. The vector q_i lies in the real span of the vectors $x_j - x_i$, so the vanishing of all A_{ij} forces $q_i = 0$. The argument works in every dimension, including dimension one. Moreover,

$$q_i = \sum_{k \neq i} \frac{m_k(x_k - x_i)}{r_{ik}^3} + M(x_i - x_{\text{cm}}).$$

Thus $q_i = 0$ is exactly the normalized central-configuration equation, and the converse follows from the same identity. The determinants vanish because every four points in the plane are affinely dependent. $\square$

The equations are scalar-product equations, rather than area-factor equations that vanish identically on a line. The equivalence is needed only for real Euclidean distance tuples; an arbitrary complex zero of F need not have a physical interpretation.

Each ordered or symmetric polynomial is homogeneous of degree one in the masses, while the determinants do not involve masses. We may therefore divide all masses by $c > 0$ and take $c = 1$. Write $k_0 = \mathbb{Q}(a, b)$, $K = \overline{k_0}$, with trivial valuation, and $X = V(F) \subset (K^*)^{10}$.

Fixed potential.

The distance equations encode centrality after the normalization $\lambda = M$, but they do not by themselves control the potential. We therefore fix that potential when applying the pointedness argument. Here $M = \sum_i m_i$; if $I = \sum_i m_i |x_i - x_{\text{cm}}|^2$, then $MI = \sum_{i < j} m_i m_j r_{ij}^2$, and the virial

identity is $U = MI$, with $U = \sum_{i < j} m_i m_j / r_{ij}$. Define

$$G = \sum_{i < j} m_i m_j r_{ij}^2 - U,$$

$$H = \sum_{i < j} m_i m_j \prod_{(k,l) \neq (i,j)} r_{kl} - U \prod_{k < l} r_{kl}.$$

Here U is an additional torus coordinate. For a fixed $u \neq 0$, write G_u, H_u for $U = u$ and put $X'_u = V(F, G_u, H_u)$. Every real normalized central configuration of potential u lies in X'_u . Lemma 9 will reduce the physical problem to finitely many such nonzero potential values. The final pointedness argument applies to these augmented varieties; it does not assert finiteness of the full complex variety $V(F)$.

Prevariety fan.

We use the support data to build a computable over-approximation of the tropical variety. Let Σ be the polyhedral fan computed by `gfan_tropicalintersection` from the supports of $f_1, \dots, f_{35}$ (max convention; file `data/fan.out`). It has f-vector $(1, 1603, 6110, 6525, 2200)$, 16,438 nonzero cones and trivial lineality space. Thus $\text{Trop}(X) \subseteq \bigcap_i \text{Trop}(V(f_i)) = |\Sigma|$.

3. GEOMETRIC AND SPECIALIZATION LEMMAS

The previous section supplies a finite support fan, but a cone in that fan is only a candidate for the tropical variety. The lemmas below explain how exact identities, specialization, and a half-space check turn that finite list of candidates into a finiteness proof.

Throughout, $\text{in}_w f$ is the max-initial form: the sum of the terms of f whose exponents maximize $\langle w, \cdot \rangle$. $\text{Trop}(X)$ is the set of $w \in \mathbb{R}^{10}$ such that $\text{in}_w I(X)$ contains no monomial. By the Fundamental Theorem [4, Thm. 3.2.3] and the Structure Theorem [4, Thm. 3.3.5], this is the tropicalization over the trivially valued algebraically closed field K . We use the max convention throughout; the min convention in [4] transfers by $w \mapsto -w$. For a cone $C = \text{cone}(v_1, \dots, v_s) \in \Sigma$ we put $w_C = \sum_j v_j$.

Lemma 3 (Cell constancy). *Let f be a polynomial whose initial form is used, with cone rays extended by a zero U -coordinate when necessary, and let*

$$\mathcal{F}_f = \bigcap_j \arg \max_{m \in \text{supp } f} \langle v_j, m \rangle.$$

If $\mathcal{F}_f \neq \emptyset$, then $\text{in}_w f = \text{in}_{w_C} f$ for every w in the relative interior of C .

Proof. Write $w = \sum \lambda_j v_j$ with all $\lambda_j > 0$. For $m_0 \in \mathcal{F}_f$ and $m \in \text{supp } f$, each $\langle v_j, m_0 - m \rangle \geq 0$, so $\langle w, m_0 - m \rangle \geq 0$. Equality holds exactly when every summand vanishes, that is, exactly when $m \in \mathcal{F}_f$. $\square$

The verifier checks the hypothesis of Lemma 3 for every generator that appears in a certificate.

Lemma 4 (Exclusion by a unit certificate). *Let Φ be a polynomial system in n torus coordinates r , over $\mathbb{Q}[a, b]$, and let $C \subset \mathbb{R}^n$ be a rational polyhedral cone. Choose $\mathcal{P} \subseteq \Phi$ satisfying Lemma 3 on C . Let $\mathcal{H}$ be the common homogeneity space of the forms $\text{in}_{w_C} f$, $f \in \mathcal{P}$, and let $\mathcal{D} \subseteq \{1, \dots, n\}$ be a coordinate index set such that $\mathcal{H} \rightarrow \mathbb{Q}^{\mathcal{D}}$ is surjective. Denote by $\bar{f}$ the form obtained by fixing $r_j = 1$ for $j \in \mathcal{D}$. Suppose*

$$\sum_{f \in \mathcal{P}} h_f \bar{f} + h_0 \left(1 - \zeta \prod_{j \notin \mathcal{D}} r_j\right) = \ell, \quad 0 \neq \ell \in \mathbb{Q}[a, b],$$

where all cofactors are in $\mathbb{Q}[a, b][\langle r_j \rangle_{j \notin \mathcal{D}}, \zeta]$. Then $\text{relint}(C) \cap \text{Trop}(V(\Phi)) = \emptyset$ over K . The identity remains an exclusion after specializing masses whenever $\ell \neq 0$ and the used initial supports are preserved.

Proof. If the initial ideal at $w \in \text{relint}(C)$ contains no monomial, its subideal generated by the chosen initial forms is proper in $K[r^{\pm 1}]$. The Nullstellensatz gives a common torus zero z . Integer vectors in $\mathcal{H}$ act on these zeros by coordinatewise torus scaling. The surjective projection onto $\mathcal{D}$ supplies an integer matrix with nonzero determinant on those coordinates. Over the algebraically closed field K , the corresponding isogeny of tori is surjective, so a zero can be moved to the slice $r_j = 1$, $j \in \mathcal{D}$. Substituting this zero and $\zeta = (\prod_{j \in \mathcal{D}} r_j)^{-1}$ into the identity gives $0 = \ell$, a contradiction. The same identity proves the specialized assertion. $\square$

We apply this lemma both to the ten distance coordinates and to the eleven coordinates (r, U) , extending each distance weight by a zero U -coordinate. The auxiliary variable ζ is distinct from the fixed potential value u .

Lemma 5 (Spreading exclusions to mass parameters). *Consider finitely many initial systems over $\mathbb{Q}[a, b]$, each with no torus zero over $\mathbb{Q}(a, b)$. There is a nonzero polynomial $q \in \mathbb{Q}[a, b]$ such that all these systems have no torus zero after any characteristic-zero specialization with $q(a, b) \neq 0$. A single such q can also preserve every support of a specified finite polynomial system.*

Proof. For each initial system, add $1 - \zeta \prod r_j$ and apply the Nullstellensatz. The resulting ideal is the unit ideal over $\mathbb{Q}(a, b)$, and hence over $\mathbb{Q}(a, b)$ by faithful scalar extension. Clearing the finitely many mass denominators in a unit identity gives a nonzero polynomial $q_i \in \mathbb{Q}[a, b]$. The product of these q_i and all nonzero coefficient polynomials of the specified system is a suitable q . When $q \neq 0$, the identities specialize and no coefficient vanishes, so the supports are unchanged. Equivalently, the parameter set on which a torus zero exists is constructible, misses the generic point, and therefore has proper closure; the identities provide a direct proof of this properness. $\square$

For augmented systems, the explicit certificates have coefficients in $\mathbb{Q}(a, b)$ and treat U as a torus variable, whereas the structural and modular exclusions use U -free forms. Applying the lemma to these systems, with U included as a coordinate when needed, produces a mass polynomial independent of u . Thus one open set works for every fixed $u \neq 0$. The coefficients of G_u, H_u also preserve their distance supports for positive masses and $u \neq 0$.

One modular fiber at the target cone is not enough: after specialization, a generic torus zero may escape toward a neighboring cone. The next lemma packages the star check that rules out this escape and makes a single finite-field computation a certificate over the generic coefficient field.

Lemma 6 (Exclusion from one modular specialization). *Let $\Phi = F \cup \{G, H\}$ as defined in Section 2, with U the extra coordinate of weight 0. For a cone $D \in \Sigma$ let Φ_D be the set of $f \in \Phi$ that satisfy Lemma 3 on D and whose form $\text{in}_{w_D} f$ does not involve U . Let $C \in \Sigma$ with $F \subseteq \Phi_C$. Suppose that there are a prime p and $(a_0, b_0) \in \mathbb{Z}^2$ such that*

- (1) *every coefficient (an element of $\mathbb{Z}[a, b]$) of every form $\text{in}_{w_C} f$, $f \in \Phi_C$, is nonzero modulo p at (a_0, b_0) , and*
- (2) *for every cone $D \in \Sigma$ having C as a face (including $D = C$), the forms $\text{in}_{w_D} f$, $f \in \Phi_C \cap \Phi_D$, specialized at (a_0, b_0) and reduced modulo p , have no common zero in the torus over $\mathbb{F}_p$.*

Then the forms $\text{in}_{w_C} f$, $f \in \Phi_C$, have no common zero in the torus over $K = \overline{\mathbb{Q}(a, b)}$. Consequently, the cone is excluded for every augmented fixed-potential variety X'_u , $u \neq 0$, at generic masses.

Proof. Write $h \in \mathbb{Z}[a, b]$ as $h = \sum \gamma_{ij}(a - a_0)^i(b - b_0)^j$ with $\gamma_{ij} \in \mathbb{Z}$, and put $\nu(h) = \min\{v_p(\gamma_{ij}) + i\sqrt{2} + j\sqrt{3}\}$. Since $1, \sqrt{2}, \sqrt{3}$ are linearly independent over $\mathbb{Q}$, the minimum is attained by exactly one term, and this term is multiplicative. So ν extends to a valuation of $\mathbb{Q}(a, b)$. Its residue field is $\mathbb{F}_p$, and $\nu(h) = 0$ iff $h(a_0, b_0) \not\equiv 0 \pmod{p}$. Extend ν to K (residue field $\mathbb{F}_p$, value group $\Gamma \subseteq \mathbb{R}$ divisible), and fix a splitting $\lambda \mapsto t^\lambda$ of ν on Γ [4, Lemma 2.1.15].

Suppose that z is a common torus zero over K , and write $z = t^{-w}y$ with $w \in \Gamma^{11}$ and $\nu(y_i) = 0$. Because the forms do not involve U , we may replace the U -entry of w by 0. For $g = \text{in}_{w_C} f =$

$\sum c_m r^m$ with $f \in \Phi_C$, condition (1) gives $\nu(c_m) = 0$ for every m . Multiplying $g(z) = 0$ by $t^{h_{\max}}$, where $h_{\max} = \max_m \langle w, m \rangle$, and reducing gives $\overline{\text{in}_w g}(\bar{y}) = 0$. Here the bar denotes specialization at (a_0, b_0) modulo p . For sufficiently small $\varepsilon > 0$, uniformly for the finitely many f , we have $\text{in}_w \text{in}_{w_C} f = \text{in}_{w'} f$ with $w' = w_C + \varepsilon w$. If $w' \notin |\Sigma|$, then some $f \in F \subseteq \Phi_C$ has a monomial $\text{in}_{w'} f$. Its specialized coefficient is nonzero, so it cannot vanish at $\bar{y}$, a contradiction. Otherwise $w' \in \text{relint } D$ for some $D \in \Sigma$. Since ε is small and $w_C \in \text{relint } C$, C is a face of D . For $f \in \Phi_C \cap \Phi_D$, Lemma 3 gives $\text{in}_{w'} f = \text{in}_{w_D} f$. Thus $\bar{y}$ is a common torus zero of the specialized forms $\text{in}_{w_D} f$, $f \in \Phi_C \cap \Phi_D$, contradicting (2). The final statement follows from the unit ideal over k_0 and Lemma 5; its generators do not involve U , so the mass condition is uniform in u . $\square$

The valuation and initial-degeneration principles behind Lemma 6 are standard; see [17, Theorem 5.6] and [4]. The lemma packages them into a criterion suited to this computation: it turns a modular calculation into a proof without any computation over $\mathbb{Q}(a, b)$. What is required is that no zero can escape through the boundary of the cell modulo p , and condition (2) expresses exactly that requirement by excluding every cone in the star of C . The modular cofactor identities and exact elimination replays are checked by `verifier/verify_lemS.py`; the target-cell replays and their exceptional branches are described in Section 5.

Remark 7 (Why the star condition is needed). The need for the star condition is visible in the following example. Emptiness of a specialized fiber can coexist with generic torus solutions even when every coefficient survives. For example, $f_1 = x + y$ and $f_2 = x + (1 + a)y + 1$ have no common zero at $a = 0$, but have the torus solution $(x, y) = (1/a, -1/a)$ for $a \neq 0$. At the neighboring weight $(1, 1)$, the specialized max-initial forms both become $x + y$, which has torus zeros. The additional star check detects this escape to infinity.

Lemma 8 (Residual half-space implies finiteness). *Let $W \subset (K^*)^n$ be an algebraic variety over a trivially valued algebraically closed field of characteristic zero. Let Σ be a finite polyhedral fan with $\text{Trop}(W) \subseteq |\Sigma|$. Let $\mathcal{E}$ be a collection of nonzero cones such that $\text{relint}(E) \cap \text{Trop}(W) = \emptyset$ for every $E \in \mathcal{E}$. Suppose each cone of $\Sigma \setminus \mathcal{E}$ is generated by its listed rays, and let $\mathcal{R}$ be the rays of these remaining cones. If $\eta \in \mathbb{Z}^n$ satisfies $\langle \eta, v \rangle > 0$ for every $v \in \mathcal{R}$, then W is finite.*

Proof. Every nonzero point of a remaining cone is a nonnegative combination of its rays, so $\langle \eta, w \rangle > 0$ for every nonzero $w \in \text{Trop}(W)$. Let Y be an irreducible component and consider the monomial map $r \mapsto r^\eta$. The tropicalization of the closure of its image is the linear image $\langle \eta, \text{Trop}(Y) \rangle$ [4, Corollary 3.2.13]. A closed irreducible subset of the one-dimensional torus is either a point or the whole torus, with tropicalization $\{0\}$ or $\mathbb{R}$, respectively. The latter cannot be contained in $[0, \infty)$, so $\langle \eta, w \rangle = 0$ on $\text{Trop}(Y)$ and hence $\text{Trop}(Y) = \{0\}$. The dimension theorem [4, Theorem 3.3.5] gives $\dim Y = 0$, and there are finitely many components. $\square$

We next isolate the fact that lets us pass from the algebraic finiteness statement at fixed potential to the original real problem.

Lemma 9 (Finitely many physical potential values). *Fix positive real masses. Among collision-free real planar central configurations normalized by $\lambda = M$, the potential takes only finitely many values, all positive. Every such configuration satisfies $U = MI = \sum_{i < j} m_i m_j r_{ij}^2$.*

Proof. Work over $\mathbb{C}$ on the center-of-mass-zero position space, restricted to configurations with all squared mutual distances nonzero. Adjoin nonzero r_{ij} satisfying $r_{ij}^2 = (x_i - x_j) \cdot (x_i - x_j)$. This distance cover is smooth and étale over the open position space because each derivative $2r_{ij}$ is invertible. The physical configurations lie on its positive real distance branches.

On this smooth cover put $f = U + \frac{M}{2}I$. Differentiating U in the position directions shows that the normalized force equations are exactly $df = 0$. The center-of-mass constraint loses no equation: the weighted sum of the vectors $\sum_{j \neq i} m_j(x_j - x_i)/r_{ij}^3 + Mx_i$ is zero. Therefore a critical point on

the constrained space satisfies the full force equations, and the normalized central configurations belong to the algebraic critical locus of f .

On every irreducible component of that locus, the differential of the restricted function f vanishes on the dense smooth locus. In characteristic zero, f is therefore constant. Dotting the force equations with $m_i x_i$ and summing gives $U = MI$, so $f = \frac{3}{2}U$ there. The critical locus has finitely many irreducible components and hence only finitely many potential values on its physical points. Finally, $MI = \sum_{i < j} m_i m_j r_{ij}^2$, and positivity gives $U > 0$. This is the physical version of [1, Lemma 2]. $\square$

Lemma 10 (The scaling ray and other negative directions). *Fix positive real masses, let $u \in \mathbb{C}^*$ and let $G_u = \sum_{i < k} m_i m_k r_{ik}^2 - u$. Put $X_u = V(F \cup \{G_u\}) \subseteq (\mathbb{C}^*)^{10}$. Every real planar central configuration lies in some X_{u_j} with $u_j \neq 0$. If $w \in \mathbb{R}^{10}$ has all coordinates negative, then $\text{in}_w G_u = -u$ is a unit, so $w \notin \text{Trop}(X_u)$. In particular the scaling ray $\mathbb{R}_{>0}(-1, \dots, -1)$ (cell 79) is excluded for every X_u . More generally, a cone on whose relative interior $\text{in}_w G_u$ is a single fixed monomial is excluded (96 cells before overlap with other exclusions).*

Proof. For a real configuration, $U > 0$, so $u_j = U \neq 0$. If all coordinates of w are negative, the monomials of G_u have w -weights $2w_{ik} < 0$, while the constant has weight 0; hence the constant is the unique initial term and is a unit. The same observation applies on any cone where the initial form is one fixed monomial. $\square$

Remark 11 (Certificates for the augmented system). Some certificates are written for $Y = V(F \cup \{G\}) \subseteq (\mathbb{C}^*)^{11}$, where $G = \sum_{i < k} m_i m_k r_{ik}^2 - U$ and U is an extra torus coordinate. For $u \neq 0$, the map $r \mapsto (r, u)$ embeds X_u into Y . Because the valuation is trivial, the U -coordinate has valuation 0, and therefore $\text{Trop}(X_u) \times \{0\} \subseteq \text{Trop}(Y)$. It follows that if a unit certificate (Lemma 4, with Lemma 3 applied in $\mathbb{R}^{11}$) shows $(w, 0) \notin \text{Trop}(Y)$ for every $w \in \text{relint } C$, then $\text{relint } C \cap \text{Trop}(X_u) = \emptyset$ for every $u \neq 0$. In these checks the cone rays receive a 0 in the U -coordinate, and initial-form constancy is verified on $C \times \{0\}$. Such certificates are therefore as useful as certificates for X in the proof below, which only uses the X_{u_j} .

Remark 12 (Certificates using the definition of U). Some certificates use both equations defining the fixed potential. Let

$$\begin{aligned} H &= \sum_{i < k} m_i m_k \prod_{(j,l) \neq (i,k)} r_{jl} - U \prod_{j < l} r_{jl} \\ &= \left(\sum_{i < k} \frac{m_i m_k}{r_{ik}} - U \right) \prod_{j < l} r_{jl}, \end{aligned}$$

which is the denominator-cleared potential identity (file `verifier/verify.py`, `augmentH`; index 36). For $u \neq 0$ put $X'_u = V(F \cup \{G_u, H_u\}) \subseteq X_u$. A real planar central configuration with potential $U = u$ satisfies $H_u = 0$, because $U = \sum m_i m_k / r_{ik}$ is its potential. It therefore lies in X'_u , and the real central configurations lie in $\bigcup_j X'_{u_j}$. As in Remark 11, $\text{Trop}(X'_u) \times \{0\} \subseteq \text{Trop}(Y')$ with $Y' = V(F \cup \{G, H\}) \subseteq (\mathbb{C}^*)^{11}$, and $\text{Trop}(X'_u) \subseteq \text{Trop}(X_u)$. Consequently, certificates for F , for $F \cup \{G\}$, and for $F \cup \{G, H\}$ (header `system F+G+H`, directory `exact/hcert/`) all exclude their cones from $\text{Trop}(X'_u)$, and the proof goes through with X'_{u_j} in place of X_{u_j} .

Proof of Theorem 1 from the certificates. We first work at generic masses over $\overline{\mathbb{Q}(a, b)}$. The identities, elimination replays, and Lemma 6 exclude the certified cones for every X'_u , $u \neq 0$. In particular, the forms used by Lemma 6 do not involve U , so the exclusion is independent of the chosen nonzero potential value.

Lemma 5, together with the denominators of the explicit identities and the coefficient polynomials of F , gives a hypersurface $B = V(q) \subset \mathbb{C}^2$, with $0 \neq q \in \mathbb{Q}[a, b]$. Outside B , the support fan and all exclusions remain valid uniformly in $u \neq 0$.

The coverage and half-space checks in Section 4 therefore make every X'_u finite by Lemma 8. Lemma 9 supplies finitely many positive potential values for the real central configurations, so their distance tuples lie in a finite union of these varieties. A labeled real distance tuple determines a configuration up to Euclidean isometry, including reflection, and $\lambda = M$ fixes scale. Finally, a nonzero polynomial cannot vanish on the entire positive quadrant, so $\mathbb{R}_{>0}^2 \setminus B$ is nonempty. Pulling it back under $(a, b, c) \mapsto (a/c, b/c)$ and clearing powers of c gives $\mathcal{O} = \{(a, b, c) \in \mathbb{R}_{>0}^3 : Q_*(a, b, c) \neq 0\}$ for a nonzero homogeneous $Q_* \in \mathbb{Q}[a, b, c]$. Thus $\mathcal{O}$ is the positive-mass complement of a hypersurface defined over $\mathbb{Q}$. □

Remark 13. Explicit Laurent certificates record mass denominators. For exclusions obtained only by structural elimination or Lemma 6, Lemma 5 proves that additional mass polynomials exist, but they are not all extracted. Consequently, the modular point $(3, 7)$ verifies the specified stars, not every exclusion at that named mass pair. An exact scan of 1,662 recorded cofactor and split-tree files finds 506 distinct denominator expressions that vanish at $(3, 7)$, spread over ten files; see `data/mass_denominator_audit.json`. Those recorded identities cannot be specialized there as written. The scan includes every archived file, including alternative identities, whereas Table 2 counts only representatives assigned to the disjoint Laurent row. It also does not cover the fourteen companion batches, whose recorded factors $q(a, b, c)$ must additionally be evaluated at $c = 1$. None of this establishes failure of finiteness at that point: a named instance would require replacement identities and point-specific replay of the structural, elimination, and modular exclusions. Avoiding the recorded denominators alone is therefore insufficient.

4. COMPUTATION

This section describes how the finite fan is screened, certified, and checked for complete coverage. The computation is organized so that the cone count, the algebraic exclusions, and the final half-space test can each be replayed independently.

Symmetry.

The swaps $(1\ 2)$ and $(3\ 4)$ preserve both the polynomial system and the fan, giving 4,781 orbits under this order-four group of the 16,438 nonzero cones. The relabeling $\sigma = (1\ 3)(2\ 4)$, together with $a \leftrightarrow b$, supplies an additional symmetry, and the generated group has order eight. The scripts `symcheck.py` and `symcheck2.py` reconstruct and check the polynomial and fan actions; every transported identity is checked again against its target cone.

Search and exact certificates.

We first screened initial systems at mass values modulo 32003. This inexpensive unit test only selected useful subsets and computation orders; it was not used as the final certificate. The direct characteristic-zero pipeline, using SINGULAR [19], then computed unit ideals over $\mathbb{Q}(a, b)$, extracted lift cofactors, and checked the resulting identities independently. Several computations became smaller after dividing initial forms by their common monomial, using a power of the product of the torus coordinates on the right-hand side, or halving exponents of variables that occur only evenly. Other cases were handled by split trees or structural elimination replays. Section 5 describes these formats and their mathematical justification.

The companion ledger.

The companion ledger keeps the large batch of related exclusions auditable: it records exactly which generators, weights, shifts, and mass factors support each identity. Fourteen Laurent-identity batches in `companion/results/` record generator indices, cell weights, monomial shifts, multipliers,

and a nonzero mass factor q . Each identity has the form

$$\sum_j c_j r^{-e_j} \text{in}_{w_C} f_{i_j} = q(a, b, c) r^d.$$

The checker expands both sides exactly, verifies that the recorded weight is the sum of the cell rays, and checks support constancy on the cell. The right-hand side is a Laurent unit at generic masses. The union also contains a structural ratio-elimination proof for cell 346 and its symmetry images, whose positive-mass branches are checked by exact Sturm arithmetic; conditional transfers from cell 325, excluded away from the mass equalities $a = 4b$ or $b = 4a$; and coefficient-level symmetry transports. The replay reconstructs the union of these proofs and verifies that it is exactly the 10,965-cell ledger. Overlapping batches are counted only once. The enumerated cell assignments are in `data/cone_accounting.json`; the ledger is checked evidence, not a list of search successes.

The companion ledger also contains a structural proof for one representative whose Laurent identity is not the most transparent description. We state its exclusion separately because it supplies both a generic complex argument and a positive-mass check.

Lemma 14 (Complex exclusion for the companion structural cell). *The initial-generator system for cell 346 has no common torus zero over $\overline{\mathbb{Q}(a, b, c)}$. It also has no common complex torus zero for any fixed positive real masses a, b, c .*

Proof. Eliminating the distance ratios $t = r_{25}/r_{15}$ and $h = r_{12}/r_{15}$ gives the necessary mass condition

$$(a - 4c)(a + 4c)(a^2 + 4c^2)^2(a^4 + 4a^3c + 6a^2c^2 + 3ac^3 + c^4)c^8P(a/c) = 0,$$

where $P(z) = z^8 - z^7 + z^6 - z^4 + 8z^3 + 3z^2 - 3z + 1$. This is a nonzero polynomial, so it excludes complex torus zeros at generic masses without any positivity assumption. For positive masses, exact Sturm arithmetic proves $P(z) > 0$ for $z > 0$; it counts roots in the *mass ratio*, not roots of the distance equations. Thus only $a = 4c$ remains. The algebraic distance equations then force $t = 1$, $h = -2$, and the necessary condition $(b^2 - 256c^2)(b^2 + 32c^2)^2 = 0$. Positivity leaves $b = 16c$. On this branch, polynomial reduction over $\mathbb{Q}$ forces $r_{14} = r_{13}$ and $r_{34} = -2r_{13}$, while another initial generator becomes $-72cr_{13}^4r_{35}^2$, a nonzero torus monomial. Distance coordinates are allowed to be complex throughout this elimination. The same exclusion transfers to the listed symmetry images, so the generic assertion satisfies the complex exclusion hypothesis of Lemma 8. $\square$

The final three exclusions.

Direct lifting remained expensive for representatives 81, 121, and 206, so each uses Lemma 6 at $(a, b) = (3, 7)$, $p = 32003$. Each star has 60 cones, including its target. All 37 initial forms of $F \cup \{G, H\}$ are constant, U -free, and retain every coefficient at this specialization. The neighboring-cone checks use modular cofactor identities, except for cluster cones 2174, 2885, and 3494, which use a short exact elimination replay. The target-cell checks use the linear even-variable reduction and the bivariate leaf argument described below. The mass-swapping symmetry also excludes their partners 316, 50, and 220.

Coverage and residual.

Table 1 gives disjoint counts by assigning each cone to the first applicable certificate in the checked ledger. Symmetry images under the pair swaps are included in their source rows, while the additional transport row records the remaining four mass-exchange images. The 96 single-term virial cells overlap earlier rows, leaving 57 in that row and the separate scaling ray. The displayed multiplicities use the order-four group generated by the two pair swaps. Mass-exchange partners are listed separately in the elimination and modular rows; the transport row counts only additional images not already assigned to an earlier row. Within each row, pair swaps give orbits of sizes one, two, or four. The enumerated accounting records every representative, target cell, orbit size, and

TABLE 1. Disjoint accounting of the nonzero support-fan cones.

Certificate or residual class	Cones
Companion Laurent and structural identities	10,965
Other Laurent identities	5,258
Resultant specialization	8
Split trees	6
Structural elimination	25
Modular star criterion	16
Additional mass-exchange transport	4
Single-term virial initial forms	57
Scaling ray	1
Excluded total	16,340
Residual	98
Total	16,438

TABLE 2. Certificate representatives and expansion for Table 1.

Class	Representatives and check in Section 5	Cones
Companion	Fourteen batches, structural and conditional transfers; companion ledger above	10,965
Laurent	1,562 representatives in <code>cert</code> , <code>dcert</code> , <code>gcert</code> , <code>hcert</code> , <code>ecert</code> ; unit identities	5,258
Resultant	227, 2440; two-variable resultant specialization	8
Split trees	378, 427; binary split identities	6
Elimination	334, 1767; 49, 2171, 2174, 2175; mass-exchange partners 2345, 218, 2885, 3494, 3545	25
Modular stars	81, 121, 206; partners 316, 50, 220	16
Transport	Remaining mass-exchange images from 353, 3097	4
Single-term G	23 representatives; enumerated accounting and Lemma 10	57
Scaling ray	79; Lemma 10	1

extra mass exchange, with preceding rows taking priority. In particular, the two split representatives contribute $2 + 4$ cones, the two resultants $4 + 4$, the elimination representatives contribute $334 : 1$; $49, 218, 1767, 2174, 2175, 2345, 3494, 3545 : 2$ each; and $2171, 2885 : 4$ each, while the six modular representatives contribute $2 + 4 + 2 + 2 + 4 + 2$.

The residual set of 98 cones is invariant under the mass-exchange symmetry and the two pair swaps; this is checked directly against the enumerated cone permutation. It has 77 distinct primitive rays, and every one has positive pairing with

$$(3) \quad \eta = (-23, 1, 1, 51, 1, 1, 51, 24, 13, 13).$$

The checker reconstructs the residual from the certificate coverage and verifies these pairings in integer arithmetic.

Interpretation and a control.

A weight w records possible distance growth orders $r_{ij} \sim t^{-w_{ij}}$. The residual cones are candidate asymptotic distance patterns compatible with the supports and not removed by our certificates; they are not claimed to be actual degenerations of real configurations. Their common half-space is enough for the finiteness argument, because no positive-dimensional tropical variety can be

supported there. The elementary example following Lemma 6 provides a negative control for the modular argument: its target specialization is empty, but the adjacent cell has torus zeros. The checker in `verifier/review_checks.py` confirms both facts and explains why a test only over $\mathbb{F}_p$ is insufficient.

5. CERTIFICATES AND INDEPENDENT VERIFICATION

An exclusion is useful only when its algebraic identity and its cone assumptions can be checked independently. This section describes the certificate formats, the exact replay, and the audits that verify the support fan and the residual coverage.

Certificate format.

`data/exact/cert/k.cof` contains:

- the cone index k ;
- the subset $\mathcal{P}$ (0-based indices into `data/sys.json`);
- the ring variables (kept coordinates and the auxiliary variable, named `u` in the files);
- for each generator in order, the terms of its cofactor C_g as
`numerator|denominator|exponent vector`,
 where numerator and denominator are in $\mathbb{Z}[a, b]$.

Verifier.

The independent checker `verifier/verify.py` uses python-flint's FLINT exact multivariate arithmetic [20]; it does not use SINGULAR or sympy. It

- (1) parses `fan.out` and `sys.json` itself;
- (2) checks the hypothesis of Lemma 3 for every $g \in \mathcal{P}$;
- (3) recomputes $\mathcal{H}$ and checks surjectivity onto the fixed coordinates $\mathcal{D}$;
- (4) clears the denominators into $\ell = \text{lcm}$ and verifies the following identity in exact arithmetic:

$$\sum (\ell C_g) \bar{g} + (\ell C_0) (1 - \zeta \prod r) = \ell.$$

The verifier also accepts the following variants.

- **system F+G**: the certificate refers to the augmented system $F \cup \{G\}$ in the 11 coordinates (r_{ij}, U) , where $G = \sum_{i < j} m_i m_j r_{ij}^2 - U$ has index 35. U has weight 0 because its fixed value is a nonzero constant with trivial valuation (Remark 11). The verifier constructs G from the masses $(a, a, b, b, 1)$. This format is used for the scaling cone 79, for the 96 cones on which $\text{in}_w G$ is a single term (a one-term certificate $C_G = 1/\text{coeff}$), and for representatives whose unit test over $\mathbb{Q}(a, b)$ timed out without G .
- **divmono**: before checking the identity, each $\bar{g}$ is divided by the greatest common monomial of its terms. Monomials are units in the Laurent ring, so this does not change the torus zero set.
- **even $v_1, \dots$** : each listed variable occurs in every $\bar{g}$ only with even exponents, and the verifier halves those exponents, i.e. writes $\bar{g}(r) = \bar{g}'(r_v^2)$. The map $r_v \mapsto s_v = r_v^2$ is a surjective endomorphism of the torus. A Laurent identity $\sum C'_i \bar{g}'_i = s^N$ therefore pulls back to $\sum C'_i(r^2) \bar{g}_i = r^{2N}$, which is again a unit certificate for $\langle \bar{g} \rangle$.
- **rhs prod^N**: no Rabinowitsch variable is used. The identity checked is $\sum_g (\ell C_g) \bar{g} = \ell \cdot \left(\prod_{j \in \mathcal{I}} r_j \right)^N$ in $\mathbb{Q}[a, b][r_{\mathcal{I}}]$, where $\mathcal{I}$ is the set of kept coordinates. On the torus the right-hand side vanishes nowhere (for $\ell(a, b) \neq 0$), so $V(\text{in}_w F) \cap T = \emptyset$, exactly as in the Rabinowitsch form.

The denominators in the certificates must lie in $\mathbb{Z}[a, b]$, and the verifier checks this. After multiplying by ℓ , each identity therefore holds in $\mathbb{Q}[a, b][r, \zeta]$. It specializes to every mass pair with $\ell(a, b) \neq 0$,

a nonempty Zariski-open (generic) condition; the finitely many polynomials ℓ are recorded in the verifier output.

Split-tree certificates.

For a few cones, a single lift identity is too large to compute even though the ideal splits naturally. Typically, one initial form factors as $g_1 g_2$ modulo a binomial of the system. For cone 378, for example,

$$a^2 + ab r_{13}^2 + ab r_{24}^2 + b^2 r_{34}^2 \equiv (a + b r_{13}^2)(a + b r_{24}^2)$$

modulo $r_{34}^2 - r_{13}^2 r_{24}^2$. A split-tree certificate (`exact/split/<k>.split`, checked by `verifier/verify_split.py`) is a finite binary tree. A node v with path ideal $\mathfrak{J}_v = \mathfrak{J} + (p_1, \dots, p_s)$ (the base Rabinowitsch ideal $\mathfrak{J}$ of `verify.py` together with the polynomials chosen at the ancestors) is either a *leaf*, carrying cofactors with $\sum_i C_i k_i = \ell_v \in \mathbb{Q}[a, b] \setminus \{0\}$, or a *split* node with polynomials q_1, q_2 and cofactors with $\sum_i C_i k_i = \ell_v q_1 q_2$. Both children, $\mathfrak{J}_v + (q_1)$ and $\mathfrak{J}_v + (q_2)$, must be certified. Over $\mathbb{Q}(a, b)$, if $q_1 q_2 \in \mathfrak{J}_v$ and $1 = x_1 + c_1 q_1 = x_2 + c_2 q_2$ with $x_j \in \mathfrak{J}_v$, then

$$1 = (x_1 + c_1 q_1)(x_2 + c_2 q_2) \in \mathfrak{J}_v.$$

Induction then gives $1 \in \mathfrak{J}$, exactly the statement certified by a `.cof` file. The support and torus-slice checks are shared with `verify.py` and are run on the header. All identities are checked exactly with `python-flint`; the generic condition is that no ℓ_v vanishes.

Elimination certificates (cones 334, 1767 and 49, 2171, 2174, 2175).

These cones are certified by explicit eliminations that the checkers `verifier/verify_334.py` and `verifier/verify_49.py` replay in exact arithmetic (`python-flint`, with no `SINGULAR`). In both groups, six initial forms $g_1, \dots, g_6$ in five variables suffice, and the forms are identical across the cones in each group. The checkers reconstruct them from the fan through the support and torus-slice checks of `verify.py`. The argument is pointwise: let z be a common torus zero of the g_i over the algebraic closure K of the coefficient field, with the masses algebraically independent. Every derived polynomial below vanishes at z , and each branch ends in a contradiction.

- (1) Three of the forms are affine-linear in three squares. Cramer's rule gives $D r^2 - N_r$ as an explicit combination of them, with D a monomial. Replacing squares and squaring away the linear unknowns then gives division-free polynomials $E_1, \dots, E_{14}$ in two variables (x, y) .
- (2) We divide only by factors known to be nonzero at z : monomials, the masses, and the factors X_q, Y_q, S_q of N_r (nonzero because $D r^2 = N_r$ and $r \neq 0$). The remaining factors F_i , the Cramer determinant of the second linear system, are treated as separate branches $F_i(z) = 0$.
- (3) Resultants lie in the ideals of their arguments, so they vanish at z . The final non-vanishing $\text{Res}_y(R', R'') \neq 0$ is certified by specialization: at one rational parameter point modulo a prime, the relevant degrees are preserved and the specialized gcd is 1. As in `verify_res.py`, this proves that the resultant is a nonzero element of $\mathbb{Z}[a, b]$. The remaining common factors are explicit. For 49 the factor is the irreducible cubic $\phi = (ay + 1)^3 - b^2(ay^3 + 1)$. Over the field $\mathbb{L} = \mathbb{Q}(a, b)[y]/(\phi)$ the checker exhibits $\psi = -\beta_0/\beta_1$ (where $\beta_1 x + \beta_0$ is the first subresultant of X_q, Y_q). It verifies $X_q(\psi) = 0$ and $(x - \psi)^2 \mid \widehat{E}_{10}, \widehat{E}_{13}$ in $\mathbb{L}[x]$, and $\deg \gcd_{\mathbb{L}}(\widehat{E}_{10}, \widehat{E}_{13}) \leq 2$. The degree bound holds because the principal subresultant coefficient is nonzero at a specialization and Gauss's lemma applies since ϕ is primitive. Thus $x(z) = \psi$ and $X_q(z) = 0$, a contradiction.

For 334, all six forms are homogeneous of degree one in (a, b) , so we divide them by $b \neq 0$ and replace a by a/b . This is an equivalence on the mass torus and justifies $b = 1$; it is not a specialization to a possibly exceptional fiber. The diagonal $r_{23} = r_{24}$ is closed by a resultant chain in $\mathbb{Z}[a]$. Off the diagonal, the symmetry $(r_{13} r_{14})(r_{23} r_{24})$ of $g_1, \dots, g_5$ (but not of g_6 , which enters only through norms $E \cdot \sigma E$) reduces the variables to $s = r_{23} + r_{24}$ and $\pi = r_{23} r_{24}$. The only common zero left, $\pi = 3/(a^2 - 1)$ and $s = -3a/(a^2 - 1)$, has $X_q Y_q S_q = 0$. These factors agree, up to

nonzero monomial factors, with the Cramer numerators n_A, n_B, n_T in Appendix A; either vanishing statement contradicts the nonzero torus coordinates. The six exact input polynomials and the division-free definitions of the derived polynomials are given in Appendix A, which also justifies the irreducibility and specialization steps.

Dehomogenized computations.

For some cones, every generator in the chosen subset is homogeneous in the masses (a, b) because the mass torus acts on the subsystem. We compute the cofactors over $\mathbb{Q}(a)$ at $b = 1$ and then rehomogenize: if

$$\sum_i C_i(a) \bar{g}_i(a, 1) = 1$$

and $\bar{g}_i$ is homogeneous of degree d_i in (a, b) , then

$$\sum_i b^{-d_i} C_i(a/b) \bar{g}_i(a, b) = 1$$

(`exact/ecert/rehom.py`). The resulting file is an ordinary certificate over $\mathbb{Q}(a, b)$, and `verify.py` checks it without referring to this construction.

Independent support-fan verification.

The `gfan` [18] computations with versions 0.7 and 0.6.2 agree on primitive rays and cones. This agreement is a cross-version replication, not an independent implementation. The version-0.7 build record lists portability changes to memory-resource headers, integer overloads, and execution policies; the original patch diff is not retained. Our separate check does not rely on those changes.

The first audit, `verifier/fan_structure.py`, independently reconstructs every one of the 16,438 nonzero cones as a cone in the common refinement of the Newton normal fans. For each support it intersects the maximizing faces on the cone rays, obtaining a face T_i with at least two terms. The corresponding normal region is described by the equalities within T_i and the inequalities $\langle w, m_0 - m \rangle \geq 0$, $m_0 \in T_i$, for the other terms. Exact rational nullspace arithmetic computes the full solution space of the face equalities. The checker verifies that every stored ray lies in this space and that its rank equals the dimension. Thus the equality solution space is exactly the cone's span, of dimension at most four; restricting to that span hides no larger normal region. Enumerating independent sets of $d - 1$ generating rays gives every facet normal. Each facet inequality appears among the defining inequalities of the normal region, proving that the stored cone equals that region. The checker also verifies that the rays are primitive and extreme, that lineality is zero, and that every facet is a stored face: 44,998 facet incidences in total. Normal-fan cones intersect in common faces, and closure under facets gives closure under all faces. Relative interiors therefore partition the support, and the nearby-cone face relation used in Lemma 6 holds independently of `gfan`. The checker also verifies invariance of all 35 supports and of the complete stored fan under the three generating relabelings.

To certify completeness of the support, `cddlib` with GMP [21] converts the 3,120 inclusion-maximal cones to exact rational half-spaces. A rational-linear solver then asks whether a weight outside their union lies in all thirty-five tropical hypersurfaces. For each support $\mathcal{A}_i$ it introduces a maximum t_i and imposes

$$t_i \geq \langle w, m \rangle \quad (m \in \mathcal{A}_i), \quad \#\{m \in \mathcal{A}_i : t_i = \langle w, m \rangle\} \geq 2.$$

Membership in the stored cones is negated using their exact half-spaces. Homogeneity allows $\max_j |w_j| = 1$. The verified order-eight symmetry action has three coordinate orbits represented by r_{12}, r_{13}, r_{15} , so only their positive and negative normalized cases need to be checked. When r_{12} is fixed, its stabilizer can place the largest of the four cross-pair coordinates at r_{13} ; when r_{13} is fixed it can arrange $w_{12} \geq w_{34}$; and when r_{15} is fixed it can arrange $w_{13} \geq w_{14}$. These inequalities reduce the search without removing any symmetry orbit.

The solver responses are not independently checkable proof objects, so both solver implementations remain part of the trusted computing base. All six cases returned **UNSAT** independently in Z3 5.1.0 [22] and Yices 2.7.0 [23], using exact rational arithmetic. The Yices replay uses standard SMT-LIB QF_LRA: at least one term attains the maximum, and every attaining term requires a different attaining term. This Boolean encoding is independent of Z3’s pseudo-Boolean operator. The half-spaces were regenerated and matched the checked inputs exactly. Because a nonempty difference of these rational polyhedral sets has a rational point, the computation proves containment for all real weights. Thus the stored fan has exactly the prevariety support required by Lemma 6. The audit code, input hashes, checked base query, and per-case results are distributed with the verifier. As a control, it proves completeness for the three-ray tropical line of $1 + x + y$ and finds a witness when one ray is removed. A timeout or **UNKNOWN** is never accepted as a completeness result.

Modular specialization certificates.

For each of the three targets $k = 81, 121, 206$, `verify_lemS.py` computes a star superset directly from `fan.out` by selecting cones whose ray sets contain the target rays. Every true coface is included, and checking extra cones is harmless. The independently verified fan structure justifies this superset. The checker requires all 37 initial forms to be constant and U -free, verifies that every coefficient remains nonzero modulo 32003 at $(3, 7)$, and checks every cone in the star. This explicitly enforces the intersection $\Phi_C \cap \Phi_D$ required by Lemma 6. Most neighboring cones have ordinary modular cofactor identities in `lemS/cert/`. The cluster cones 2174, 2885, and 3494 instead use `cluster_mod.py`: it reconstructs six forms, derives three square relations by Cramer’s rule, replays division-free elimination to fourteen bivariate polynomials, and checks that their common roots force a zero torus coordinate. The transported cluster forms are reconstructed at their actual cones.

Target-cell replays.

The checker `verify_lemS_rep.py` reconstructs all target forms, validates the homogeneous torus slice, and replaces the three even variables by their squares. One form eliminates a squared coordinate, while a second gives

$$c_{23}(x, y)s_{23} + c_{24}(x, y)s_{24} = 0.$$

Outside $c_{23} = c_{24} = 0$, put $s_{23} = \kappa P$, $s_{24} = \kappa Q$, with κ, P, Q and the eliminated squared coordinate nonzero. The homogeneous equations then give forms independent of κ . Three of them determine S^2, T^2, W^2 , and two cube equations make $1/T$ and $1/W$ affine in $1/S$. Squaring the first relation determines $1/S$ rationally. Every denominator is checked to be nonzero on the branch; for cell 206, an additional bivariate leaf proves that the required denominator cannot vanish. Substitution into further initial forms gives four bivariate polynomials, whose common zero set is checked exactly outside the explicitly excluded factors. Cases in which a cube-relation coefficient vanishes are handled separately by four further leaves, so no branch is discarded by division.

The exceptional branch $c_{23} = c_{24} = 0$ gives $(x^3, y^3) = (1, 1)$ for 81 and 121, and $(1, -1/13)$ for 206. Since $32003 \equiv 2 \pmod{3}$, a cube root of the nonzero constant lies in $\mathbb{F}_{32003}$, and all roots lie in its quadratic extension. Each branch has nine root pairs. The generator `b0_generate.py` found small cofactor identities for these cases, and the independent checker `b0_verify.py` reconstructs their input polynomials and verifies all nine identities in $\mathbb{F}_{32003}[t]/(t^2 + t + 1)$. The branch check is mandatory; a partial replay cannot certify a cell.

Bivariate leaf check.

The helper `leafchk.py` computes a nonzero resultant in x from two of the input polynomials. Every common affine root has a y -coordinate among the roots of this resultant. For each irreducible factor, the checker forms the corresponding finite extension field, specializes all input polynomials, and computes their univariate gcd in x . It removes only factors belonging to the explicitly excluded

polynomials. A constant gcd in every fiber proves that no permitted common root exists. Extra variables are rejected. This is finite-field exact arithmetic and does not call SINGULAR.

Coverage and pointedness.

The complete replay reconstructs the cones excluded by actual identities and elimination branches, then checks (3) on every residual ray. The companion replay checks its Laurent, structural, and symmetry proofs against the 10,965-cell union. Search output and cached success markers are not treated as exclusions.

Trusted arithmetic and solvers.

The independent replay relies on Python, the verifier code, and FLINT through python-flint for exact rational, polynomial, and finite-field operations; the companion identities additionally use SymPy's exact expansion. The support-completeness audit relies on cddlib/GMP, Z3, and Yices for rational SMT solving. The recorded Z3 version is 5.1.0. Singular generates characteristic-zero identities but is not used to verify them. Gfan generates the initial fan; the normal-region and facet checks independently verify its structure, while the solver audit verifies completeness. These are executable exact checks with stated software dependencies, not a proof-assistant formalization.

6. CODE AND CERTIFICATE DATA

The source, verifier, and certificate archive are versioned together so that the computations can be replayed from a single release. They are available at

<https://github.com/brandonlign/tropcc>

The certificate archive is attached to release v1.3.0; its SHA-256 digest and all extracted file hashes are recorded in `certificate_manifest.json`. The repository README gives installation and full replay commands; `CERTIFICATES.md` maps claims to checks.

APPENDIX A. EXPLICIT ELIMINATION INPUTS AND DERIVED POLYNOMIALS

The appendix records finite algebraic definitions instead of expanding the large resultant coefficients. Together with the branch prescriptions below, these definitions determine every polynomial used in the two elimination replays. The six input forms are also stored in `data/elimination_for_ms.json`, and the independent reconstruction checks their agreement with F and the stated cones.

Cone group 334.

Put $(A, B, T, x, y) = (r_{13}, r_{14}, r_{34}, r_{23}, r_{24})$ and take $b = 1$ after the homogeneous mass reduction above. The homogeneous forms are recovered from the displayed normalized forms by $\tilde{g}_i(a, b; r) = b g_i(a/b; r)$. Thus the mass-independent coefficient terms acquire a factor b , while distance-slice constants retain their meaning. The six normalized forms, in their checked order, are

$$\begin{aligned} g_1 &= -A^3 B^3 a - A^3 \\ &\quad + AB^3 a (x - 1)(x + 1) - AB^2 \\ &\quad + AT^2 - 2B^3 \\ g_2 &= -A^3 B^3 a + A^3 B a (y - 1)(y + 1) \\ &\quad - 2A^3 - A^2 B \\ &\quad - B^3 + BT^2 \\ g_3 &= A^2 y^3 + B^2 x^3 \\ &\quad - 2ax^3 y^3 - x^3 y^2 - x^3 - x^2 y^3 - y^3 \end{aligned}$$

$$\begin{aligned}
g_4 &= A^2axy^3 + T^2x \\
&\quad - ax^3y^3 - axy^3 - x^3 - xy^2 - 2y^3 \\
g_5 &= B^2ax^3y + T^2y \\
&\quad - ax^3y^3 - ax^3y - 2x^3 - x^2y - y^3 \\
g_6 &= -2B^3T^3a - B^3T^2y \\
&\quad + B^3y(x-y)(x+y) - B^2T^3ay \\
&\quad - T^3ay(y-1)(y+1)
\end{aligned}$$

The same forms occur at cone 1767. The variables A, B, T in this appendix are local distance variables and are unrelated to A_{ij} .

Cone group 49.

Put $(A, B, T, x, y) = (r_{14}, r_{15}, r_{45}, r_{34}, r_{35})$. In this group both parameters remain. The six forms are

$$\begin{aligned}
g_1 &= -A^3B^3b - A^3 \\
&\quad + AB^3b(x-1)(x+1) - AB^2 \\
&\quad + AT^2 - 2B^3b \\
g_2 &= -A^3B^3b + A^3Bb(y-1)(y+1) \\
&\quad - 2A^3 - A^2Bb \\
&\quad - B^3b + BT^2b \\
g_3 &= A^2by^3 + B^2x^3 \\
&\quad - 2ax^3y^3 - bx^2y^3 - by^3 - x^3y^2 - x^3 \\
g_4 &= A^2axy^3 + T^2x \\
&\quad - ax^3y^3 - axy^3 - 2by^3 - x^3 - xy^2 \\
g_5 &= B^2ax^3y + T^2by \\
&\quad - ax^3y^3 - ax^3y - bx^2y - by^3 - 2x^3 \\
g_6 &= -2B^3T^3b - B^3T^2by \\
&\quad + B^3by(x-y)(x+y) - B^2T^3ay \\
&\quad - T^3ay(y-1)(y+1)
\end{aligned}$$

These forms agree at cones 49, 2171, 2174, and 2175. The two groups use different initial subsets and distance slices, which explains the different mass coefficients in g_1, g_2, g_6 . Their forms g_3, g_4, g_5 agree at $b = 1$, so the X_q, Y_q, S_q used for group 334 are exactly the $b = 1$ specializations of those defined below for group 49.

A common division-free construction.

The two elimination replays share a division-free reduction. In either group write

$$(g_3, g_4, g_5)^\top = \mathbf{M}(A^2, B^2, T^2)^\top - v,$$

where $\mathbf{M}, v$ are obtained by coefficient extraction from the displayed forms. Put $d = \det \mathbf{M}$, and let n_A, n_B, n_T be its Cramer numerators. Thus

$$d(A^2, B^2, T^2)^\top - (n_A, n_B, n_T)^\top = \text{adj}(\mathbf{M})(g_3, g_4, g_5)^\top.$$

The determinant d is a nonzero monomial and hence a unit on the generic mass and distance torus. For a polynomial P , define $\rho(P)$ termwise by replacing each $A^{2i+e}B^{2j+f}T^{2k+h}$ with

$n_A^i n_B^j n_T^k A^e B^f T^h d^{N-i-j-k}$, where $e, f, h \in \{0, 1\}$ and N is the largest $i + j + k$ among its terms. On a common zero, $\rho(P) = d^N P$. Let prim remove integer content and the greatest common monomial, including mass monomials. These removed factors are nonzero on the generic torus. Set $R_i = \text{prim } \rho(g_i)$ for $i = 1, 2, 6$, and extract coefficients by

$$R_1 = A(a_1 B + b_1) + c_1 B,$$

$$R_2 = B(a_2 A + c_2) + b_2 A,$$

$$R_6 = T(a_6 B + b_6) + c_6 B.$$

The lowercase coefficients here are polynomials in x, y over the mass ring, not mass parameters. Applying prim ρ to $c_1^2 B^2 - A^2(a_1 B + b_1)^2$, $b_2^2 A^2 - B^2(a_2 A + c_2)^2$, and $c_6^2 B^2 - T^2(a_6 B + b_6)^2$ gives, respectively,

$$\alpha B + \beta, \quad \gamma A + \delta, \quad \alpha_6 B + \beta_6.$$

Next, take $\text{Res}_A(R_1, R_2)$ and $\text{Res}_B(R_1, R_2)$, remove their maximal factors B and A , respectively, and apply prim ρ . The results are $\mu B + \omega_0$ and $\mu_2 A + \omega_2$. Define the following polynomials, applying prim to each right-hand side:

$$\begin{aligned} E_1 &= \beta^2 d - \alpha^2 n_B, & E_2 &= \delta^2 d - \gamma^2 n_A, \\ E_3 &= \alpha_6 \beta - \alpha \beta_6, \\ E_4 &= a_1 \beta \delta - b_1 \alpha \delta - c_1 \gamma \beta, \\ E_5 &= a_2 \delta \beta - c_2 \beta \gamma - b_2 \delta \alpha, \\ E_6 &= (b_1 \delta)^2 d - (c_1 \gamma - a_1 \delta)^2 n_B, \\ E_7 &= (b_2 \delta)^2 d - (c_2 \gamma - a_2 \delta)^2 n_B, \\ E_8 &= (c_1 \beta)^2 d - (b_1 \alpha - a_1 \beta)^2 n_A, \\ E_9 &= (c_2 \beta)^2 d - (b_2 \alpha - a_2 \beta)^2 n_A, \\ E_{10} &= \omega_0^2 d - \mu^2 n_B, & E_{11} &= \mu \beta - \alpha \omega_0, \\ E_{12} &= \mu \beta_6 - \alpha_6 \omega_0, & E_{13} &= \omega_2^2 d - \mu_2^2 n_A, \\ E_{14} &= \mu_2 \delta - \gamma \omega_2. \end{aligned}$$

Each E_i vanishes at a common torus zero. The constructions use multiplication, squaring, resultants, and division only by known units.

Factors and branches for group 49.

The following factors must be nonzero:

$$\begin{aligned} X_q &= a^2 x^4 y^4 + a b x^3 y^4 + a b x y^4 + b^2 y^4 - x^4, \\ Y_q &= a^2 x^4 y^4 + a x^4 y^3 + a x^4 y - b^2 y^4 + x^4, \\ S_q &= a^2 x^4 y^4 - b^2 y^4 - b x^3 y - b x y^3 - x^4. \end{aligned}$$

Indeed $n_A = -2x^3 X_q$, $n_B = -2by^3 Y_q$, $n_T = 2ax^3 y^3 S_q$. Remove their maximal powers from E_i to obtain $\tilde{E}_i$. On the leaf, also remove powers, while the quotient is nonconstant, of

$$ax^2 y^2 \pm by^2 \pm x^2 \quad (\text{all four sign choices}), \quad ax^3 y^3 + by^3 + x^3,$$

to obtain $\hat{E}_i$. The zero loci of these five factors are checked separately using $\tilde{E}_1, \tilde{E}_2, \tilde{E}_3$, so no division discards a branch. The leaf uses $\text{Res}_x(\hat{E}_{10}, \hat{E}_{13})$ and $\text{Res}_x(\hat{E}_{10}, \hat{E}_{12})$ as described in Section 5.

Irreducibility and the specialization bound.

Let $\phi = (ay+1)^3 - b^2(ay^3+1)$. Over $\mathbb{Q}(a, y)$ the quotient $(ay+1)^3/(ay^3+1)$ is not a square: its order at $y = -1/a$ is three, since the denominator there is $1 - a^{-2} \neq 0$. Hence ϕ , viewed as a quadratic

in b , is irreducible over $\mathbb{Q}(a, y)$. Its two coefficients are coprime in $\mathbb{Q}(a)[y]$, so Gauss's lemma gives irreducibility in $\mathbb{Q}(a)[y, b]$. Viewed instead as a polynomial in y , it is primitive over $\mathbb{Q}(a)[b]$ (its y -coefficient is the unit $3a$ there), and is therefore irreducible over $\mathbb{Q}(a, b)$. Thus $\mathbb{L} = \mathbb{Q}(a, b)[y]/(\phi)$ is a field.

A principal subresultant coefficient is polynomial in the coefficients of its two arguments. If it vanished in $\mathbb{L}$, its polynomial numerator would be divisible by ϕ after clearing coefficient denominators. Primitivity in $\mathbb{Z}[a, b][y]$ follows because the coefficients $3a$ and $1 - b^2$ are coprime over $\mathbb{Q}[a, b]$; Gauss's lemma then makes the divisibility valid before specialization. The coefficient would consequently vanish at every modular root of the specialized ϕ for which denominators survive. The checked specialization preserves the x -degrees and has gcd degree two, so its degree-two principal subresultant coefficient is nonzero. This proves the generic bound $\deg \gcd_{\mathbb{L}}(\widehat{E}_{10}, \widehat{E}_{13}) \leq 2$. Together with the verified double root $x = \psi$, this identifies the gcd and yields $X_q = 0$, a contradiction.

Branches for group 334.

On the diagonal, $x = y$ first gives $B = \pm A$. The negative sign forces $T = 0$. For the positive sign, put $(A, B, x, y, T) = (z, z, t, t, v)$ and remove the common monomial from each specialized g_i , obtaining P_i . Choose signs of P_1, P_4 so that their v^2 coefficients are $-1, +1$. Define

$$\begin{aligned} Q_1 &= \text{prim Res}_v(P_6, P_4), \\ Q_2 &= \text{prim Res}_z(P_1 + P_4, P_3), \\ Q_3 &= \text{prim Res}_z(Q_1, P_3). \end{aligned}$$

The exact computation gives $0 \neq \text{Res}_t(Q_2, Q_3) \in \mathbb{Z}[a]$, excluding a common zero at generic a . Off the diagonal, let $s = x + y$, $\pi = xy$, and use the swap $(A \ B)(x \ y)$. For $1 \leq i \leq 14$, $i \neq 3, 12$, take the symmetric part $E_i + \sigma E_i$ and the divided antisymmetric part $(E_i - \sigma E_i)/(x - y)$; for $i = 3, 12$, take the norm $E_i \sigma E_i$. Rewrite each expression exactly in (s, π) and verify by back-substitution. The branch factors are

$$\begin{aligned} a\pi^2 - s^2 + 2\pi, \quad a\pi^2 + s^2 - 2\pi, \\ a\pi^3 + s^3 - 3s\pi, \quad a^2\pi^4 - s^4 + 4s^2\pi. \end{aligned}$$

The checker treats their zero loci separately. On the leaf it uses the symmetric and antisymmetric parts of E_{10} and the norm of E_{12} . Write $\widehat{E}_{10s}, \widehat{E}_{10a}, \widehat{E}_{12n}$ for the leaf quotients of the symmetric, divided antisymmetric, and norm polynomials after removing only the nonzero branch factors. Put

$$R_1 = \text{Res}_s(\widehat{E}_{10s}, \widehat{E}_{10a}), \quad R_2 = \text{Res}_s(\widehat{E}_{10s}, \widehat{E}_{12n}), \quad P_* = (a^2 - 1)\pi - 3.$$

Remove the a -content and all factors π, P_* from R_1 , leaving R'_1 . Preserved degrees and the coprime specialization $a = 7$ modulo 1000003 give $\text{Res}_\pi(R'_1, R_2) \neq 0$ over $\mathbb{Q}(a)$. Hence a common zero has $\pi = 0$ or $P_* = 0$, and the torus excludes $\pi = 0$. The exact factorization further gives

$$\begin{aligned} \text{prim}_s \gcd(\text{Res}_\pi(\widehat{E}_{10s}, P_*), \text{Res}_\pi(\widehat{E}_{10a}, P_*)) &= L_*^e, \\ L_* &= (a^2 - 1)s + 3a, \end{aligned}$$

up to a unit in $\mathbb{Q}(a)$, with $e > 0$; prim_s removes parameter content. For $T_* = n_A n_B n_T$ in (s, π) , the checker verifies $L_* \mid \text{Res}_\pi(T_*, P_*)$. Because P_* is linear, $P_* = L_* = 0$ forces $T_* = 0$, which is impossible on the torus. The mass exception $a^2 = 1$ is included when spreading.

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